

KARTIK KUMBHAR

☎ +919322130400 ✉ kumbharkartik150@gmail.com 🔗 LinkedIn 🌐 GitHub

Achievements

- **Winner – Allianz Tech Championship '25**, awarded **€3,000 scholarship** for an innovative technical solution.
- **Received project funding of AU\$250** from **La Trobe University, Australia**.
- **Runner Up** Technology Grand Infusion Challenge 2025 (**International Level**)
- Secured **1st Prize** in **Prakalp 2025**, a national-level project competition held at **JSPM Tathawade**.
- **Won Domain Master Title** – Morpheus National Level Hackathon held in SIT Lonavala.
- **Won Best Research Paper award** in ETFI 2026 titled 'IoT-Enabled Smart Umbrella for Environmental Monitoring and User Convenience'
- **Pupil : CodeForces** , solved **450+ problems** across various **platforms**.

Experience

Research Project : Learning Disabilities in Children | *La Trobe University, Australia* **Jan 2025 - Jan 2026**

- Developed an AI-driven detection system **achieving 80% accuracy**, improving **performance by 70%** over traditional methods.
- **Implemented and optimized ML models** (SVM, Random Forest, Neural Networks) using Python, Scikit-learn, and Pytorch with advanced preprocessing and hyperparameter tuning, **boosting precision and recall by 18%**.
- Designed an **end-to-end ML pipeline for data ingestion, feature engineering, model training, and evaluation**, enabling automated retraining and consistent performance across datasets.

Full Stack Developer Intern | *Auriquant Designs* **Jan 2026 - Mar 2026**

- Built a **full-stack restaurant platform** using **Flutter** and **Spring Boot microservices** to manage menus, orders, billing, and reservations, delivering a scalable and modular architecture. .
- Implemented secure **RBAC authentication (Spring Security + JWT)** and **real-time order tracking via WebSockets**, improving workflow transparency and coordination across restaurant staff.
- **Reduced API response time by 40% using Redis caching** and automated ordering, billing, and analytics workflows, cutting **manual operations by 70%** and enabling data-driven decision-making.

Projects

GAII - Generative AI Impact Analytics – Real-Time CO Tracking & ESG Optimization Platform (Live) (GitHub)

- Designed GAIA, a sustainability-focused framework for Generative AI, **enabling measurable reductions in token usage (15%+) through prompt optimization** and usage monitoring.
- Implemented a **Chrome Extension + Flask backend** capable of handling real-time multi-platform AI tracking with low-latency data processing.
- Created an **ESG analytics dashboard providing actionable CO insights and reports**, designed for enterprise-scale extensibility and compliance.

HireX - AI-Powered Professional Networking Platform (GitHub)

- Built a **scalable full-stack platform** using React, Express, FastAPI, MongoDB, Redis, and Qdrant supporting networking, chat, events, and hiring workflows.
- Developed an **AI-based identity verification pipeline** with face liveness detection and **vector similarity matching** to reduce fraudulent and duplicate accounts.
- Implemented **JWT/OAuth authentication, WebSockets, QR-based event systems, and Dockerized deployment**, enabling secure and scalable real-time user experiences.

Technical Skills

Languages: Java(Proficient) , Python

Frontend Development: React

Backend Development : SpringBoot

Cloud & DevOps: Docker, AWS

Machine Learning & AI : Scikit-learn, PyTorch, Deep Learning, LLMs

Developer Tools: VS Code, Git/GitHub, Android Studio, Postman, Jupyter

Education

Dr. D.Y Patil Institute of Technology

Bachelor of Engineering : Computer Engineering GPA: 9.16/10

2023-2027

Pune, Maharashtra